



Data Structure

Level 1

Circular Linked List

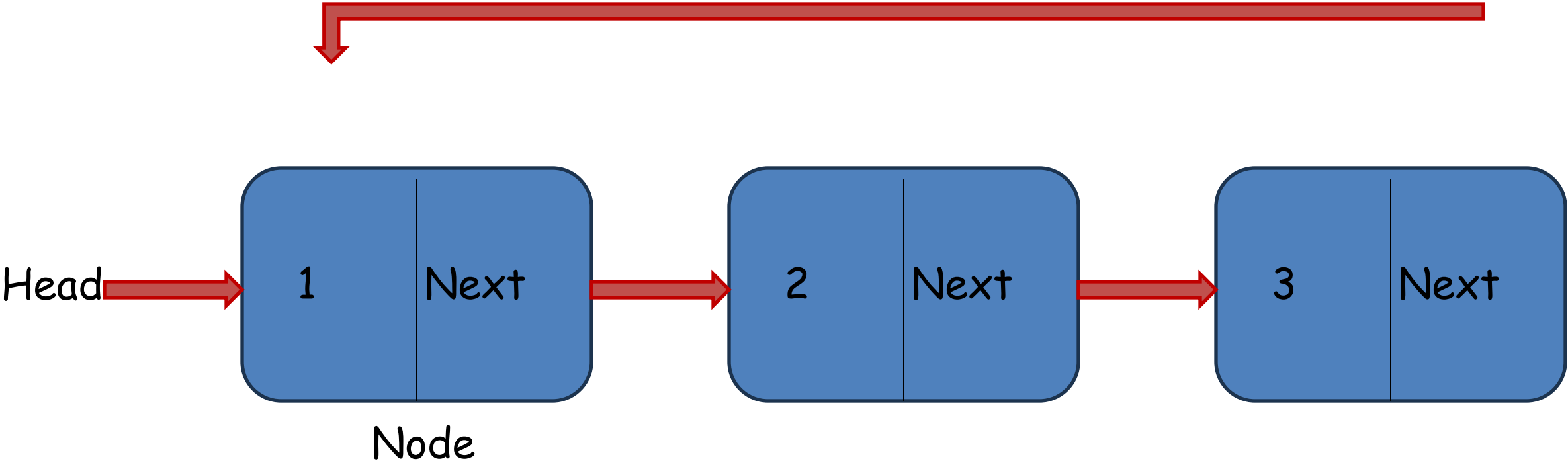


Data Structure

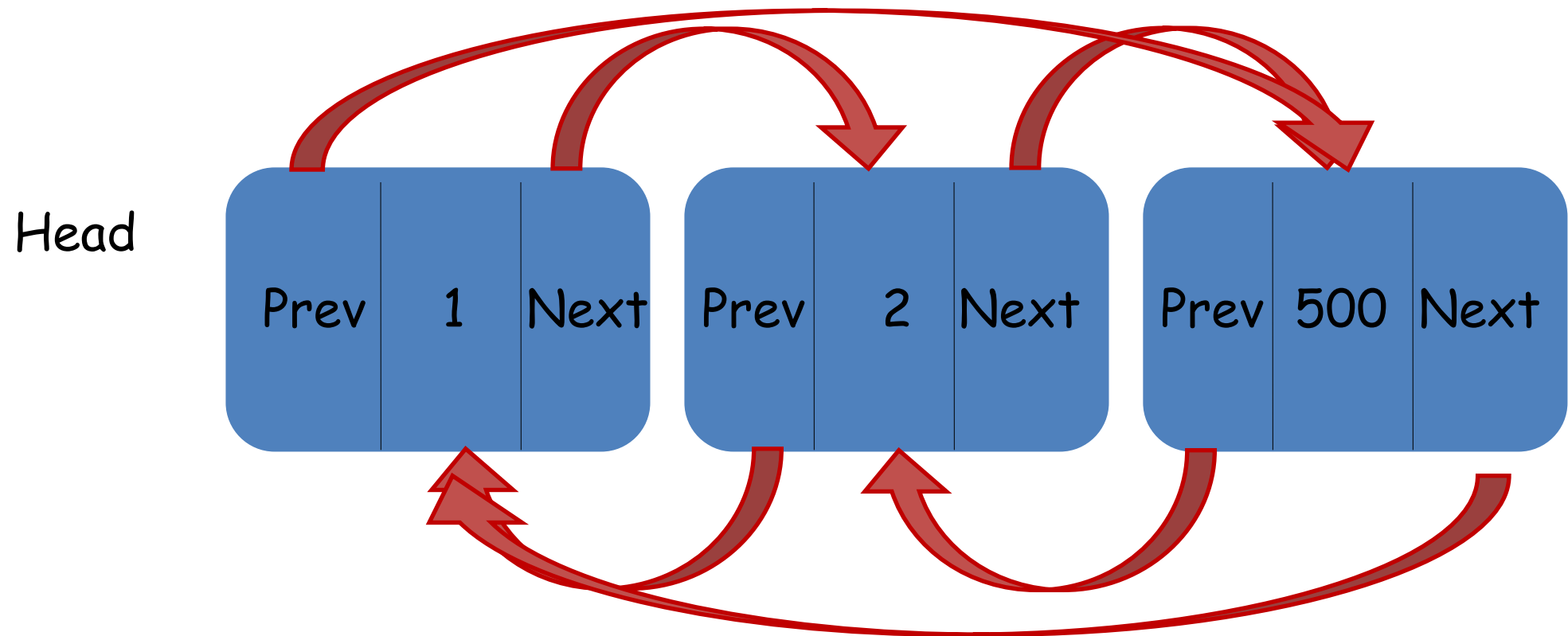
Level 1

Circular Linked List

Linked List (Circular) Data Structure



Doubley Linked List (Circular) Data Structure



Definition

A Circular Linked List is a linked list where the last node points back to the first node instead of pointing to `NULL`.

Explanation

◆ In a normal linked list:

- ▣ The last node points to `NULL`

- ▣ The list has a clear beginning and end

◆ In a Circular Linked List:

- ▣ The last node points back to the first node

- ▣ There is no `NULL` at the end

- ▣ The nodes form a circle

Procedures (Circular Linked List)

1. Create the First Node

Steps:

▨ Create a new node

▨ Store the value

▨ Make the node point to itself

➡ Result:

1



2. Add More Nodes

Steps:

▨ Create a new node

▨ Connect it to the previous node

▨ Update the last node to point to the first node

➡ Result:

1 → 2 → 3



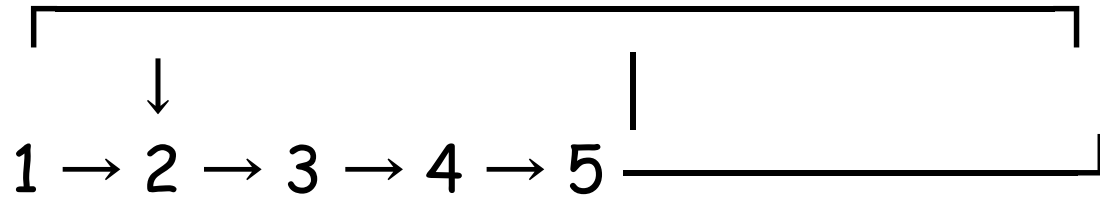
☐ Procedures (Circular Linked List)

☐ Traverse the List

☐ Steps:

- ☐ Start from the first node
- ☐ Follow the `next` pointer
- ☐ Stop when you reach the starting node again
- ☐ → Result: All nodes are visited exactly once.

💡 Example



➡ The `next` pointer of node `5` points back to node `1`.

Note

- ◆ The last node does not contain `NULL` in its `next` pointer.
- ◆ To know when traversal is complete, we usually check whether we have returned to the starting node.
- ◆ Circular Linked Lists can be:
 - ▣ Singly Circular Linked List → each node has `next`
 - ▣ Doubly Circular Linked List → each node has `next` and `prev`

Comparison

Normal Linked List

- ✓ Last node points to `NULL`
- ✓ Has a clear end
- ✓ Traversal stops at `NULL`

Circular Linked List

- ✓ Last node points to the first node
- ✓ No `NULL` at the end
- ✓ Traversal stops when we return to the starting node

⊘ Warning

⚠ A normal condition like:

`while (current != NULL)`

should not be used for a Circular Linked List because `current` will never become `NULL`.

⚠ You need another stopping condition, such as returning to the starting node.

✦ Summary

- ✓ Circular Linked List connects the last node back to the first node
- ✓ The last node does not point to NULL
- ✓ Forms a circular structure
- ✓ Traversal stops when we reach the starting node again

Thank you 😊